IDA LAB-1

This lab consists of FIVE Practical

Practical-1:

- Obtained the descriptive statistics for the given data using MSEXCEL.

Sample of daily production in meters of 30 carpet looms					
16.2	16.8	15.9	15.6	15.9	16.6
15.8	16	16	15.7	15.9	15.6
15.8	16.4	16.3	16	16.8	15.6
15.8	15.2	16	16.2	15.4	16.9
16.3	15.9	16.4	16.1	15.7	16.3

- Check the obtained results using R.

Practical-2:

- This Practical relates to the **College** data set, which can be found in the file **College.csv**. It contains a number of variables for 777 different universities and colleges in the US. The variables are

  - Private : Public/private indicator
  - Apps : Number of applications received
  - Accept : Number of applicants accepted
  - Enroll : Number of new students enrolled
  - Top10perc : New students from top 10 % of high school class
  - Top25perc : New students from top 25 % of high school class
  - F.Undergrad : Number of full-time undergraduates
  - P.Undergrad : Number of part-time undergraduates
  - Outstate : Out-of-state tuition
  - Room.Board : Room and board costs
  - Books : Estimated book costs
  - Personal : Estimated personal spending
  - PhD : Percent of faculty with Ph.D.'s
  - Terminal : Percent of faculty with terminal degree
  - S.F.Ratio : Student/faculty ratio
  - perc.alumni : Percent of alumni who donate
  - Expend : Instructional expenditure per student
  - Grad.Rate : Graduation rate

Before reading the data into R, it can be viewed in Excel or a text editor.

(a)
Use the read.csv() function to read the data into R. Call the loaded data college. Make sure that you have the directory set to the correct location for the data.

(b)
Look at the data using the fix() function. You should notice that the first column is just the name of each university. We don't really want R to treat this as data. However, it may be handy to have these names for later. Try the following commands:

```
> rownames(college)=college[,1]
> fix(college)
```

You should see that there is now a row.names column with the name of each university recorded. This means that R has given each row a name corresponding to the appropriate university. R will not try to perform calculations on the row names. However, we still need to eliminate the first column in the data where the names are stored. Try

```
> rownames(college)=college[, -1]
> fix(college)
```

Now you should see that the first data column is Private. Note that another column labeled row.names now appears before the Private column. However, this is not a data column but rather the name that R is giving to each row.

(c)

(i) Use the summary() function to produce a numerical summary of the variables in the data set.

(ii) Use the pairs() function to produce a scatterplot matrix of the first ten columns or variables of the data. Recall that you can reference the first ten columns of a matrix A using A[,1:10].

(iii) Use the plot() function to produce side-by-side boxplots of Outstate versus Private.

(iv) Create new qualitative variable, called Elite, by *binning* the Top10perc variable. We are going to divide universities into two groups based on whether or not the proportion of students coming from the top 10 % of their high school classes exceeds 50 %.

```
> Elite=rep("No",nrow(college))
> Elite[college$Top10perc>50]="Yes"
> Elite=as.factor(Elite)
> college=data.frame(college,Elite)
```

Use the summary() function to see how many elite universities there are. Now use the plot() function to produce side-by-side boxplots of Outstate versus Elite.

(v) Use the hist() function to produce some histograms with differing numbers of bins for a few of the quantitative variables. You may find the command par(mfrow=c(2,2)) useful: it will divide the print window into four regions so that four plots can be made simultaneously. Modifying the arguments to this function will divide the screen in other ways.

Practical-3:

- This Practical involves the Auto data set. Make sure that the missing values have been removed from the data.

(a)
Which of the predictors are quantitative, and which are qualitative?

(b)
What is the range of each quantitative predictor? You can answer this using the range() function.

(c)
What is the mean and standard deviation of each quantitative predictor?

(d)
Now remove the 10th through 85th observations. What is the range, mean, and standard deviation of each predictor in the subset of the data that remains?

(e)
Using the full data set, investigate the predictors graphically, using scatterplots or other tools of your choice. Create some plots highlighting the relationships among the predictors. Comment on your findings.

(f)
Suppose that we wish to predict gas mileage (mpg) on the basis of the other variables. Do your plots suggest that any of the other variables might be useful in predicting mpg? Justify your answer.

Practical-4:

- This Practical involves the Boston housing data set.

(a)

To begin, load in the Boston data set. The Boston data set is part of the MASS library in R.

> library(MASS)

Now the data set is contained in the object Boston.

> Boston

Read about the data set:

> ?Boston

How many rows are in this data set? How many columns? What do the rows and columns represent?

(b)

Make some pairwise scatterplots of the predictors (columns) in this data set. Describe your findings.

(c)

Are any of the predictors associated with per capita crime rate? If so, explain the relationship.

(d)

Do any of the suburbs of Boston appear to have particularly high crime rates? Tax rates? Pupil-teacher ratios? Comment on the range of each predictor.

(e)

How many of the suburbs in this data set bound the Charles river?

(f)

What is the median pupil-teacher ratio among the towns in this data set?

(g)

Which suburb of Boston has lowest median value of owner-occupied homes? What are the values of the other predictors for that suburb, and how do those values compare to the overall ranges for those predictors? Comment on your findings.

(h)

In this data set, how many of the suburbs average more than seven rooms per dwelling? More than eight rooms per dwelling? Comment on the suburbs that average more than eight rooms per dwelling.

Practical-5:

- This Practical involves the creation of dataset for Surya Bank Pvt. Ltd. And answering the following questions.

(a)

Refer the following for creation of Surya Bank Dataset.

SURYA BANK FINANCIAL RESULTS

Sl. No.	Financial Year	Net Profit	Total Income	Operating Expenses
1	2000	10.24	188.91	35.62
2	2001	5.37	203.28	49.03
3	2002	9.33	240.86	50.97
4	2003	14.92	258.91	97.42
5	2004	17.07	250.07	99.20
6	2005	–20.1	204.19	80.72
7	2006	10.39	237.33	86.68
8	2007	16.55	280.64	96.52
9	2008	33.01	361.51	95.23
10	2009	58.17	486.67	114.74
11	2010	23.92	625.94	194.10
12	2011	26.30	610.19	202.14

(b)
Calculate the mean, variance, coefficient of skewness and kurtosis for the different models which help in creating the customer awareness about e-banking. Compare the results of the different modes of creating
awareness (Q4).
(c)
Which of the e-banking facilities, on an average, influences the customer most while selecting the bank? (Q7)
(d)
Which facility has the highest variability (Q7).
(e)
Comment on the average satisfaction level of the customers with the e-services provided by their banks. Also calculate the variance and coefficient of skewness of the satisfaction level of the customers (Q12).